Functions: Composite, Inverse & Logarithmic

Domain and range

1 Find the domain of $f(x) = \frac{1}{x^2 - 9}$.

2 Find the domain of $g(x) = \sqrt{7 - 2x}$.

3 Find the range of $f(x) = -(x-2)^2 + 5$.

Composite functions

4 Given $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $(f \circ g)(x)$.

5 Given $f(x) = \sqrt{x+4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(x)$.

6 Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.

Inverse functions

7 Find $f^{-1}(x)$ for $f(x) = \frac{2x-5}{3}$.

8 Find $f^{-1}(x)$ for $f(x) = x^3 + 2$.

Verifying inverses

9 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses by showing $f(g(x)) = x$.

10 Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.

Logarithm laws

11 Expand $\log_b\left(\frac{x^3 y}{z^2}\right)$ using the laws of logarithms.

12 Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.

13 Evaluate $\log_2 32 - \log_2 4$ without a calculator.

Solving logarithmic equations

14 Solve $\log_4(x^2) = 3$.

15 Solve $\ln x + \ln(x - 3) = \ln 10$.

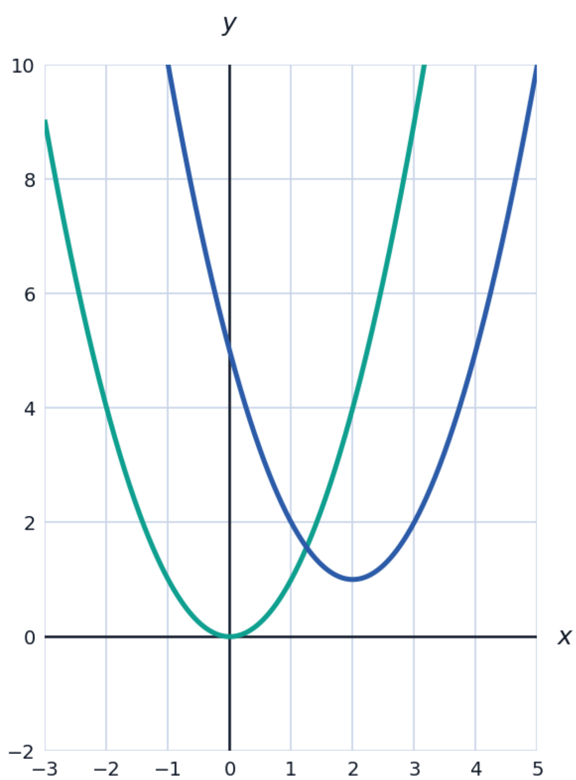
Exponential equations and logs as inverses

16 Solve $e^{2x-1} = 20$, giving your answer to three decimal places.

17 The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

Transformations of functions

18 The graph shows $f(x) = x^2$ (teal) and $g(x) = (x - 2)^2 + 1$ (indigo). Describe the transformation that maps f onto g.



19 If $h(x) = -f(x + 3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h.

A well-designed word problem: investment growth and doubling time

20 This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t, to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.